

ASSIGNMENT :2

- ❖ Secure networks with firewalls NAT, port filtering, packet filtering, and other methods.

Task (1):

New Inbound Rule Wizard

Protocol and Ports

Specify the protocols and ports to which this rule applies.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

Does this rule apply to TCP or UDP?

☒ TCP

☐ UDP

Does this rule apply to all local ports or specific local ports?

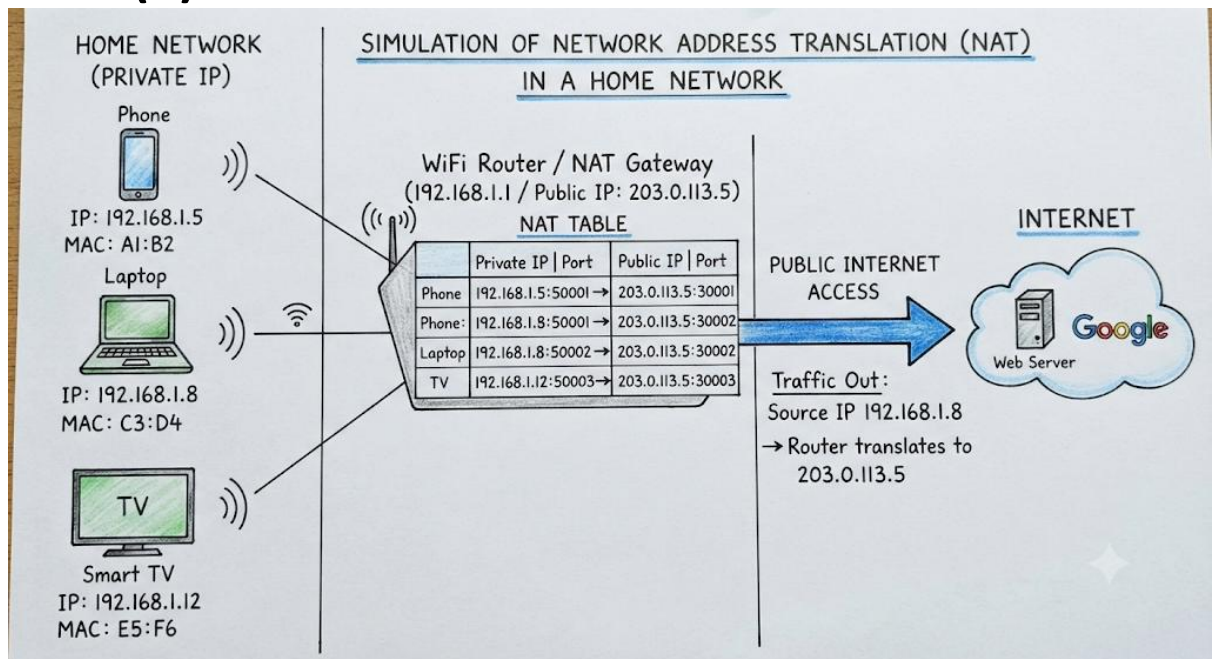
☐ All local ports

☒ Specific local ports:

Example: 80, 443, 5000-5010

< Back Next > Cancel

Task (2):



Task (3):

- ❖ To block outgoing traffic on port 25 (SMTP) using your WiFi router's admin panel, follow these steps:

1. Access the Router Admin Panel:

- Open a web browser, type your router's default gateway IP address (e.g., 192.168.1.1 or 192.168.0.1) into the address bar, and log in with your admin credentials.

2. Navigate to Firewall / Port Filtering:

- Locate the security settings, usually found under Advanced, Security, or Firewall, and select IP/Port Filtering or Access Control.

3. Create the Outbound Block Rule:

- Add a new rule with the following parameters: set direction to Outbound (or Outgoing), protocol to TCP, destination port to 25, and action to Block or Deny.

4. Save and Apply Settings:

- Click Save, Apply, or reboot the router if required to enforce the new firewall policy.

Task (5):

1. Application-Layer Protection via Web Application Firewalls (WAF):

- **Payload Inspection:** Instead of just checking IP addresses, WAFs inspect Layer 7 HTTP/HTTPS packet payloads to identify malicious patterns such as SQL injection, cross-site scripting (XSS), or malformed API requests targeting user profiles or restaurant menus.

2. Perimeter Defense and DDoS Mitigation:

- **Layer 3/4 Packet Filtering:** Network firewalls analyze incoming packet headers at the transport and network layers to filter traffic based on source IP reputation, port numbers, and abnormal connection request rates.

3. Internal Microsegmentation and Lateral Security

- **Service-to-Service Filtering:** Internal firewalls are deployed between the hundreds of microservices that power these applications (such as separating the user authentication service from the payment gateway).